

|          |                                                                                                                                                                          |           |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          |                                                                                                                                                                          |           |
| 9(a)(i)  | force is downwards/down the page<br><b>or</b><br>current is (right) to left                                                                                              | <b>B1</b> |
|          | by left-hand rule, field is into plane of paper                                                                                                                          | <b>B1</b> |
| 9(a)(ii) | magnetic force provides the centripetal force                                                                                                                            | <b>C1</b> |
|          | $Bqv = mv^2 / r$                                                                                                                                                         | <b>C1</b> |
|          | $v = Bqr / m$<br><br>$= (8.0 \times 10^{-4} \times 1.60 \times 10^{-19} \times 6.4 \times 10^{-2}) / (9.11 \times 10^{-31})$<br><br>$= 9.0 \times 10^6 \text{ m s}^{-1}$ | <b>A1</b> |
| 9(b)(i)  | arrow showing field direction down the page                                                                                                                              | <b>B1</b> |
| 9(b)(ii) | $Bqv = Eq$<br><b>or</b><br>$v = E / B$                                                                                                                                   | <b>C1</b> |
|          | $E = 9.0 \times 10^6 \times 8.0 \times 10^{-4}$<br><br>$= 7.2 \times 10^3 \text{ N C}^{-1}$                                                                              | <b>A1</b> |
| 9(c)     | straight line/undeviated                                                                                                                                                 | <b>B1</b> |
|          | condition for no deflection depends only on $v$<br><b>or</b><br>condition for no deflection does not depend on $m$ or $q$                                                | <b>B1</b> |